University of Heidelberg
Faculty of Physics and Astronomy

Begginners Physics Lab 1

Experiment 15 – Inclined Plane

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AI-Disclaimer

AI tools were used to assist in translating, as well as in the generation of text in section 1

1. Introduction

The experiment investigates the fundamental mechanics of rigid bodies moving down an inclined plane. By observing different rolling bodies specifically a solid cylinder and a hollow cylinder the experiment explores how mass distribution affects rotational and translational motion. The main objectives are to quantitatively determine the acceleration of these bodies as they roll down the ramp and to verify the law of conservation of mechanical energy by comparing the initial potential energy with the final translational and rotational kinetic energy.

2. Basic Principles and theories

The motion of a round body rolling down an inclined plane without slipping is governed by both linear and rotational dynamics. Two primary physical frameworks are utilized to analyze this motion: Newton's laws of motion and the principle of energy conservation.

2.1. Dynamics and Acceleration

When a body rolls down an inclined plane with an angle φ, it experiences a downhill force component from gravity ($F_H = mg\sin\varphi$) and an opposing static friction force (F_R). This friction force exerts a torque (moment of force) M relative to the center of mass, causing the body to rotate. The relationship between torque, the moment of inertia I, and angular acceleration $\frac{d\omega}{dt}$ is given by:

$$M = F_R \cdot r = I\frac{d\omega}{dt} \tag{2.1}$$

Using the rolling without slipping condition, the velocity of the center of gravity is $v_s = r\omega$. Differentiating this yields the relationship between linear acceleration a_s and angular acceleration, allowing us to express the friction force as:

$$F_R = \frac{I}{r^2} \cdot a_s \tag{2.2}$$

Applying Newton's second law for linear motion along the plane ($ma_s = mg\sin\varphi - F_R$) and substituting the friction force, the linear acceleration of the rolling body is derived as:

$$a_s = \frac{mg\sin\varphi}{m + \frac{I}{r^2}} \tag{2.3}$$

This formula demonstrates that the acceleration depends on the body's moment of inertia, explaining why a hollow cylinder and a solid cylinder roll at different rates despite having the same radius.[1]

2.2. Conservation of Energy

The motion can also be analyzed through the conservation of mechanical energy. Assuming no energy is lost to non-conservative forces (like air resistance or kinetic friction, since the static friction does no work), the initial potential energy is fully converted into kinetic

energy. The total energy W_{ges} is the sum of translational kinetic energy, rotational kinetic energy, and potential energy:

$$W_{ges} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + mgh \tag{2.4}$$

As the body descends a vertical height h, the decrease in potential energy (mgh) equals the increase in total kinetic energy ($\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$).

2.3. Gaussian error propagation

The errors of the measured quantities are propagated using the Gaussian error propagation formula:

$$\sigma_f = \sqrt{\sum_{i=1}^{N} \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_{x_i}^2} \tag{2.5}$$

as well as the addition of statistical and systematic errors in quadrature:

$$\sigma_{total} = \sqrt{\sigma_{stat}^2 + \sigma_{sys}^2} \tag{2.6}$$

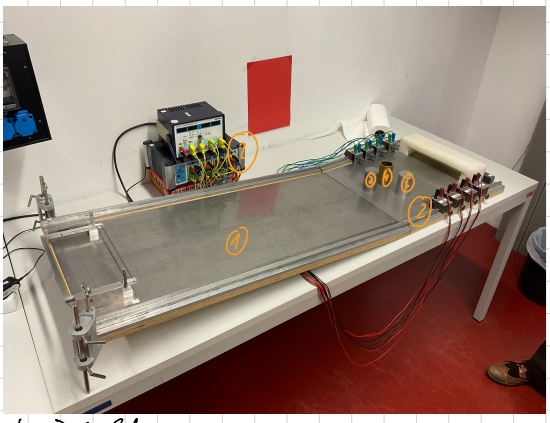
3. Measuring Protocol

3.1. Setup and Tasks

Protocoll: Experiment 15: Inclined Plane

Jonathan Gießen, Theo Semmler 15.9.2026

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Img 3.1 Setup

Setup :

- Height-adjustable runway (1)
- Light-barrier with stopwatch (2)
- Spirit level
- solid cylinder: alu, $\rho = 2{,}70\ \frac{g}{cm^3}$ (3)
- hollow cylinder brass, $\rho = 8{,}44\ \frac{g}{cm^3}$ (4)
- coposite cylinder : sheath of aluminum, core of brass (5)
- caliper
- Scale

A1: Measurment

$\Delta\sigma = \frac{\sigma}{\sqrt{n}}$ $\Delta c = 0{,}05\,cm$

We adjust for the error of the caliper not being upright by doing multiple measurements.

Total error: $\Delta = \sqrt{\Delta\sigma^2 + \Delta c^2}$ with $\Delta\sigma = 0{,}37$

Tbl 3.1: Measurment of rolling bodies

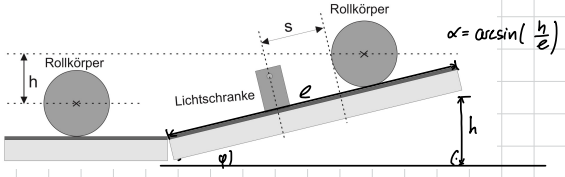
cylinder	d_{inner} [mm]	d_{outer} [mm]	mass [g]
solid	0	5,00±0,05	443,0 ± 1,0
hollow	4,65 ±0,05	5,00±0,05	443,0 ± 1,0
(composite)	0,76±0,05	5,00±0,05	443,0± 1,0

Height of track [cm]:
1: 11,10
2: 12,80
3: 11,50
4: 12,80
5: 12,70

$\bar{h} = (12{,}2 \pm 0{,}4)\,cm$

length of track [cm]: 100,0 ± 0,5cm

With the error estimation based on the error of the ruler and the uncertainty of the measurement method the plane the ruler is set on is not perfectly flat and there is a tiny gap between the measured length and the actual plane (round)



The time error for all time measurements is $\Delta t = \pm 0{,}001\,s$

A2: movement type? consistent acceleration

$L_i = L_{end} - L_{start}$, $\Delta ruler = 0{,}05\,cm$

Tbl 3.2 Full cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25{,}00 \pm 0{,}07$	$t_1 = 0{,}821$
$l_2 = 36{,}00 \pm 0{,}07$	$t_2 = 1{,}084$
$l_3 = 49{,}00 \pm 0{,}07$	$t_3 = 1{,}272$
$l_4 = 64{,}00 \pm 0{,}07$	$t_4 = 1{,}451$

Hollow cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25{,}00 \pm 0{,}07$	$t_1 = 1{,}011$
$l_2 = 36{,}00 \pm 0{,}07$	$t_2 = 1{,}209$
$l_3 = 49{,}00 \pm 0{,}07$	$t_3 = 1{,}403$
$l_4 = 64{,}00 \pm 0{,}07$	$t_4 = 1{,}604$

Composed cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25{,}00 \pm 0{,}07$	$t_1 = 0{,}855$
$l_2 = 36{,}00 \pm 0{,}07$	$t_2 = 1{,}018$
$l_3 = 49{,}00 \pm 0{,}07$	$t_3 = 1{,}186$
$l_4 = 64{,}00 \pm 0{,}07$	$t_4 = 1{,}356$

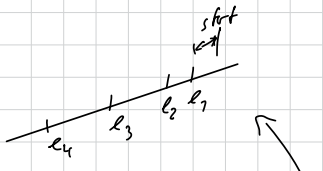
The uncertainty of the light source not being perfectly centered is neglected in the face of the much bigger uncertainty of the ruler. Nevertheless it is worth for a mention.

We aligned both sides (sender and receiver) with the ruler so no alignment error aside from Δr is expected.

We add the two separate Δr:

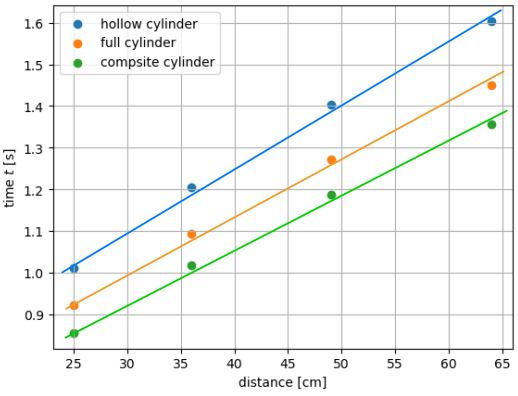
$\Delta r_{tot} = \sqrt{\Delta r_1^2 + \Delta r_2^2} = \sqrt{2} \cdot 0{,}05 =$

Δr_1: sender side, Δr_2: receiver side



$s = \frac{1}{2} a t^2 \Rightarrow t \sim \sqrt{s}$

Plots:



A3 Tbl: 3.3 Measurment of acceleration

[cm] dist Δs_{tof}=0,07	time hollow Δt = [ms]					t full [ms]				
l_1 = 25	1,019	1,027	1,014	1,022	1,009	0,922	0,914	0,900	0,901	0,913
l_2 = 36	1,212	1,223	1,209	1,215	1,201	1,096	1,089	1,075	1,074	1,087
l_3 = 49	1,410	1,426	1,410	1,415	1,400	1,278	1,268	1,256	1,254	1,266
l_4 = 64	1,610	1,629	1,613	1,614	1,600	1,458	1,446	1,435	1,435	1,446

Table 3.4

You didn't define the standard error so far

Mean + standard error:

	Hollow [s]	Full [s]
C_1:	1,018 ± 0,004	0,906 ± 0,006
C_2:	1,212 ± 0,004	1,080 ± 0,005
C_3:	1,412 ± 0,004	1,260 ± 0,006
C_4:	1,613 ± 0,005	1,440 ± 0,006

A4:

:
```
# messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])
```

Tbl. 3.5 Measurement of translational velocity Hollow:

[cm] Distance Δx_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L = 25{,}00 \pm 0{,}07$	0,270	0,271	0,270	0,270	0,269

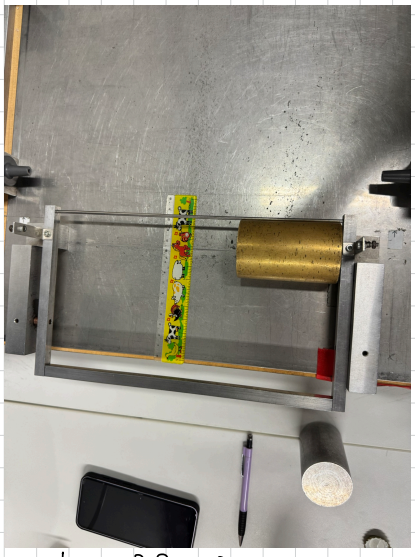
$\overline{T_H} = (0{,}2700 \pm 0{,}0010)$s with $\Delta\overline{T_H} = \sqrt{\left(\frac{\sigma_{T_H}}{\sqrt{5}}\right)^2 + \Delta t^2}$ $\Rightarrow \overline{v_H} = \frac{\Delta L}{\overline{T_H}}$

Tbl. 3.6 Measurement of translational velocity Full:

[cm] Distance Δx_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L = 25{,}00 \pm 0{,}07$	0,243	0,244	0,243	0,244	0,243

Error propagation!

$\overline{T_F} = (0{,}2430 \pm 0{,}0010)$s with $\Delta\overline{T_F} = \sqrt{\left(\frac{\sigma_{T_F}}{\sqrt{5}}\right)^2 + \Delta t^2}$ $\Rightarrow \overline{v_F} = \frac{\Delta L}{\overline{T_F}}$



Img 3.2 Distance



$h = \sin(\alpha) \cdot (\ell - 11\text{mm})$

3.2. Execution of the Experiment

The Experimental procedure involves the following main steps:

1. **Setup and Geometry Measurement:** Familiarizing with the light barrier control unit and stopwatch system. Measuring the diameters of the test bodies (solid and hollow cylinders) using a caliper. Using a spirit level to ensure the inclined plane has an even inclination across its entire width, adjusting the supports if necessary (No adjustment was needed in this case).

2. **Qualitative Analysis of Motion:** Verifing uniformly accelerated motion by placing the light barriers at appropriate intervals ($t \propto \sqrt{s}$). Observing and explain qualitatively why bodies with different mass distributions require different amounts of time to reach the bottom.

3. **Quantitative Acceleration Measurement:** Place the light barriers along the incline and measure their exact distances s from the starting mechanism. Release the solid and hollow cylinders and record the time it takes for them to pass the barriers. Repeat this measurement 5 times for each rolling body.

4. **Energy Conservation Study:** To evaluate the conservation of energy, two light barriers are posiioned on the horizontal section at the end of the ramp. The final translational velocity of the bodies are measured. The calculated total kinetic energy will then be compared to the initial potential energy, which is derived from the height difference of the plane.

4. Evaluation

4.1. Exercise 1: Comparison of Acceleration (theoretical vs. experimental)

The acceleration of the rolling bodies can be calculated using the derived formula:

$$a_s = \frac{g \sin \varphi}{1 + \frac{I}{mr^2}} \tag{4.1}$$

The Moment of inertia I for a solid cylinder is $I = \frac{1}{2}mr^2$, and for a hollow cylinder, it is $I = \frac{1}{2}m(r_{out}^2 + r_{in}^2)$
With the measured parameters of the cylinders, the theoretical accelerations can be calculated. The parameters are summarized in Table 4.1.

Table 4.1: Parameters of the cylinders used in the experiment.

Parameter	**Solid Cylinder**	**Hollow Cylinder**
Mass [g]	443.0 ± 1.0	443.0 ± 1.0
Inner diameter [m]	0	$(46.50 \pm 0.05) \times 10^{-3}$
Outer diameter [m]	$(50.00 \pm 0.05) \times 10^{-3}$	$(50.00 \pm 0.05) \times 10^{-3}$
Moment of Inertia [kg $\times$ m^2]	$(1.384 \pm 0.004) \times 10^{-4}$	$(2.582 \pm 0.007) \times 10^{-4}$

To calculate the theoretical acceleration, the angle of inclination φ of the ramp is needed. This can be determined from the height difference h and the length of the ramp L using the relation:

$$\sin\varphi = \frac{h}{L} \tag{4.2}$$

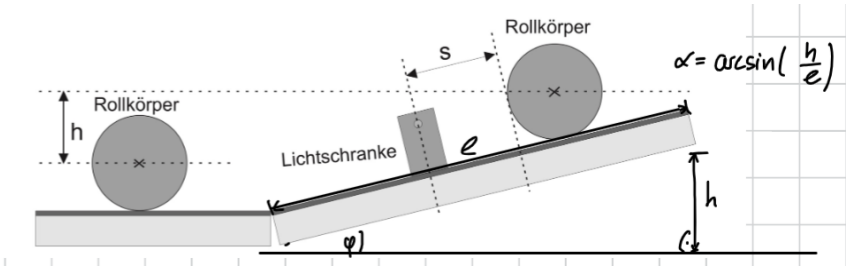


Figure 4.1: Schematic representation of the inclined plane with height h and length L. [1]

with the measured height $h = (10.1 \pm 0.4) \times 10^{-2}\,\mathrm{m}$[1] and the length of the track $l = (1.000 \pm 0.005)\,\mathrm{m}$, the angle of inclination can be calculated as:

$$\varphi = \arcsin\left(\frac{h}{L}\right) = \arcsin\left(\frac{(10.1 \pm 0.4) \times 10^{-2}}{1.000 \pm 0.005}\right) \tag{4.3}$$

$$\approx 5.797 \pm 0.024^\circ = (0.1012 \pm 0.0004)\,\mathrm{rad} \tag{4.4}$$

You have included an extra zero here, but I think for the next calculations you use the correct uncertainty
-1

Therefore, the theoretical accelerations for the solid and hollow cylinders can be calculated as (Equation 2.3):

$$a_{solid} = \frac{g\sin\varphi}{1 + \frac{I_{solid}}{mr^2}} = (0.661 \pm 0.026)\,\frac{\mathrm{m}}{\mathrm{s}^2} \tag{4.5}$$

$$a_{hollow} = \frac{g\sin\varphi}{1 + \frac{I_{hollow}}{mr^2}} = (0.513 \pm 0.021)\,\frac{\mathrm{m}}{\mathrm{s}^2} \tag{4.6}$$

Now, the experimentally determined times are plotted quadratically t^2 against the distance s. The slope of the linear fit corresponds to the acceleration, which can be compared to the theoretical values.

Where do you get the uncertainties for these plotted points from?
-1

[1] The height was measured two days later, because the initial measurement was not accurate.

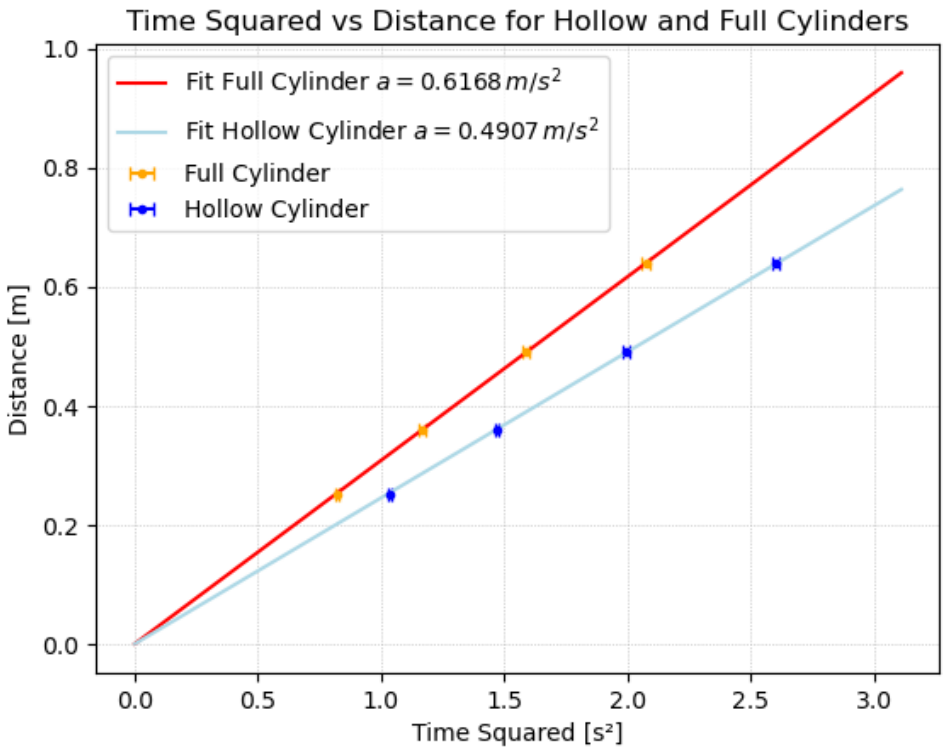


Figure 4.2: Comparison of theoretical and experimental accelerations for solid and hollow cylinders.

The function used for the linear fit is $t^2 = \frac{2}{a}s$, where a is the acceleration. The slope of the fit line gives the value of $\frac{2}{a}$. The accelerations obtained from the fit are:

How do you perform the fit and extract these uncertainty values? Your model forces the line through (0,0), is this affecting the gradient?
-1

$$a_{\text{solid, exp}} = (0.617 \pm 0.004)\,\frac{\text{m}}{\text{s}^2} \tag{4.7}$$

$$a_{\text{hollow, exp}} = (0.4907 \pm 0.0027)\,\frac{\text{m}}{\text{s}^2} \tag{4.8}$$

The experimental results show that the solid cylinder accelerates faster than the hollow cylinder, which is consistent with the theoretical predictions.
Interestingly, the experimental acceleration of both cylinders is much lower than the theoretical values.
The sigma-deviation of the experimental values from the theoretical ones can be calculated as:

$$\sigma_{\text{solid}} = \frac{|a_{\text{solid, exp}} - a_{\text{solid, theo}}|}{\sigma_{\text{hollow, theo}}} \Rightarrow 1.73\sigma \tag{4.9}$$

$$\sigma_{\text{hollow}} = \frac{|a_{\text{hollow, exp}} - a_{\text{hollow, theo}}|}{\sigma_{\text{solid, theo}}} \Rightarrow 1.19\sigma \tag{4.10}$$

here you use sigma to represent two different values in the same equation, which is very confusing. You also calculate this value in the next table and get a different value this is presumably more correct, since these values (4.9 and 4.10) don't take into account the uncertainty in the experimental values.

Conclusion

The experimental results for the accelerations of the solid and hollow cylinders show a significant deviation from the theoretical predictions, with both results being lower than expected. The calculated values are:

Value	**Solid Cylinder**	**Hollow Cylinder**
Theoretical Acceleration:	0.661 ± 0.026	0.513 ± 0.021
Experimental Acceleration:	0.617 ± 0.004	0.4907 ± 0.0027
Deviation:	$1.69\ \sigma$	1.06σ

4.2. Exercise 2: Calculating the kinetic Energy

For calculatiung the kinetic energy, the final velocities of the cylinders are measured using two light barriers at the end of the ramp. The time taken to pass between the two barriers is recorded, and the velocity is calculated as:

$$v = \frac{d}{t} \tag{4.11}$$

Through the conservation of energy, the initial potential energy $E_{pot} = mgh$ should equal the total kinetic energy $E_{kin} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. The rotational kinetic energy can be expressed in terms of the linear velocity using the rolling condition $\omega = \frac{v}{r}$.
The height difference equation can be derived from the geometry and the starting distance from the end of the ramp.

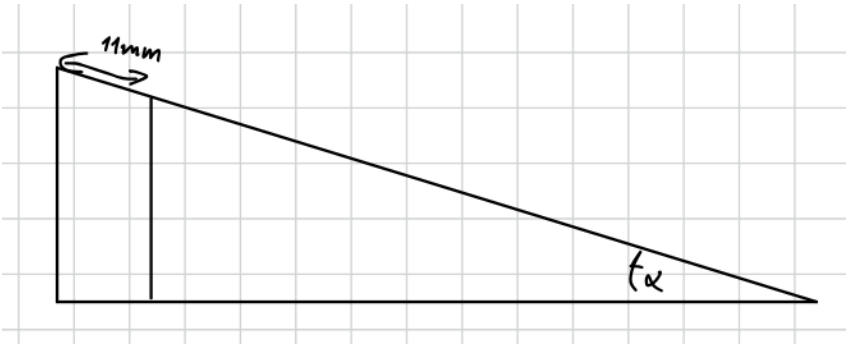


Figure 4.3: Schematic representation of the height difference h and the length of the ramp L.

For the height difference h, we can use the following relation (intercept theorem):

$$h_{eff} = \frac{h}{l} \cdot (l - 11\text{mm}) = (0.100 \pm 0.004)\,\text{m} \tag{4.12}$$

The potential energy for the cylinders equals:

$$E_{pot} = (0.443 \pm 0.001)\,\text{kg} \cdot 9.81\,\frac{\text{m}}{\text{s}^2} \cdot (0.100 \pm 0.004)\,\text{m} = (0.434 \pm 0.018)\,\text{J} \tag{4.13}$$

The measured times and speeds for the solid and hollow cylinders with $r = (25.000 \pm 0.025) \times 10^{-3}\,\text{m}$ to pass between the two light barriers with a distance of $d = (25.00 \pm 0.07) \times 10^{-2}\,\text{m}$ are:

Table 4.2: Measured times and calculated speeds for the solid and hollow cylinders.

Cylinder Type	**Time [s]**	**Speed [m/s]**	ω **[rad/s]**
Solid Cylinder	0.2430 ± 0.0010	1.0271 ± 0.0051	41.08 ± 0.21
Hollow Cylinder	0.2700 ± 0.0010	0.9259 ± 0.0043	37.04 ± 0.18

You need to show the error propagation for the values in tables.
-1

The kinetic energies for the solid and hollow cylinders can be calculated as follows:

$$E_{kin} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \tag{4}$$

This results in the following kinetic energies for the solid and hollow cylinders:

Table 4.3: Calculated kinetic energies for the solid and hollow cylinders.

Cylinder Type	**Translational KE [J]**	**Rotational KE [J]**	**Total KE [J]**
Solid Cylinder	0.2337 ± 0.0024	0.1168 ± 0.0012	0.351 ± 0.004
Hollow Cylinder	0.1900 ± 0.0019	0.1771 ± 0.0018	0.367 ± 0.004

The translational and rotational energies have not been defined, only kinetic energy.
-1

Comparing the total kinetic energies with the initial potential energy, we get:

$$E_{kin,solid} = (0.351 \pm 0.004)\,\text{J} \tag{4.15}$$
$$E_{kin,hollow} = (0.367 \pm 0.004)\,\text{J} \tag{4.16}$$
$$E_{pot} = (0.434 \pm 0.018)\,\text{J} \tag{4.17}$$

This results in a deviation of:

$$\Delta E_{solid} = E_{pot} - E_{kin,solid} = (0.083 \pm 0.018)\,\text{J} \tag{4.18}$$
$$\Rightarrow 4.61\sigma \tag{4.19}$$
$$\Delta E_{hollow} = E_{pot} - E_{kin,hollow} = (0.067 \pm 0.018)\,\text{J} \tag{4.20}$$
$$\Rightarrow 3.72\sigma \tag{4.21}$$

Conclusion

The total kinetic energies for both cylinders are lower than the initial potential energy, indicating that some energy was lost during the rolling motion:

Value	**Solid Cylinder**	**Hollow Cylinder**
Initial Potential Energy:	0.434 ± 0.018	0.434 ± 0.018
Total Kinetic Energy:	0.351 ± 0.004	0.367 ± 0.004
Deviation:	$4.61\ \sigma$	3.72σ

5. Discussion of Results

In figure 4.2 you can see that the acceleration of the solid cylinder is higher than that of the hollow cylinder, which is consistent with the theoretical predictions. This is due to

the fact that the solid cylinder has a lower moment of inertia, allowing it to accelerate more quickly down the incline.
Even though the experimental results show the expected trend, there are significant deviations from the theoretical values. The experimental accelerations for both cylinders are lower than the theoretical predictions. The kinetic energies show deviations from the theoretical predictions as well. The lower experimental accelerations suggest that additional resistive forces, such as rolling friction may have influenced the motion of the cylinders. The total kinetic energies for both cylinders are lower than the initial potential energy, indicating that some energy was lost during the rolling motion, likely due to friction.
Since friction is proportional to the normal force, which is in turn proportional to the mass of the rolling body, it is expected that the solid cylinder would experience a greater frictional force than the hollow cylinder. This could explain why the solid cylinder has a lower kinetic energy than the hollow cylinder, despite having a higher acceleration, as well as bigger deviations from the theoretical predictions, which neglect resistive forces. The solid cylinder's greater mass results in a larger normal force, leading to increased frictional losses during its motion down the incline.

They are the same mass...

-1

It is very unlikely that the timing and measurement errors are the main contributors to the observed deviations, as the experimental setup was carefully calibrated and repeated measurements were taken to minimize random errors. I suggest that the measurement of the height difference should be repeated, as it was measured two days later and could have been affected by systematic errors, because the height difference that, was measured in the beginnung was a lot higher than the one measured two days later. Identifying and measuring the height of the slope was not very precise, because holding the caliper perpendicularly to the surface was difficult. Determining the geometry of the ramp more accurately could help to reduce the discrepancies between the theoretical and experimental results.

It is possible that there are measurment errors that are systematic rather than "random"

6. Conclusion

All in all, the experiment successfully demonstrated the principles of rolling motion and energy conservation, while also highlighting the importance of considering real-world factors such as friction and measurement accuracy in experimental physics. It was a lot of fun to do this experiment, and I learned a lot about the physics of rolling motion and the importance of careful measurement and analysis in experimental physics. I also learned how to use the precise light barrier system and the stopwatch system, which was very interesting.

Total minus points: -6
Grade: +

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References

[1] Dr. J.Wagner, *Physikalisches Anfängerpraktikum*, Stand 11/2024

[2] W. Ilberg, M. Krötzsch, *Physikalisches Praktikum für Anfänger*, BSB B. G. Teubner Verlagsgesellschaft, Leipzig (1985).

A. Code

height of track

[81]:
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
```

[82]:
```
h = np.array([11.1, 12.8, 11.5, 12.8, 12.7])
err_h = 0.05

print(np.mean(h))
error_mean = np.std(h, ddof=1)/np.sqrt(len(h))
print(error_mean)
print(np.sqrt(error_mean**2 + err_h**2))
```

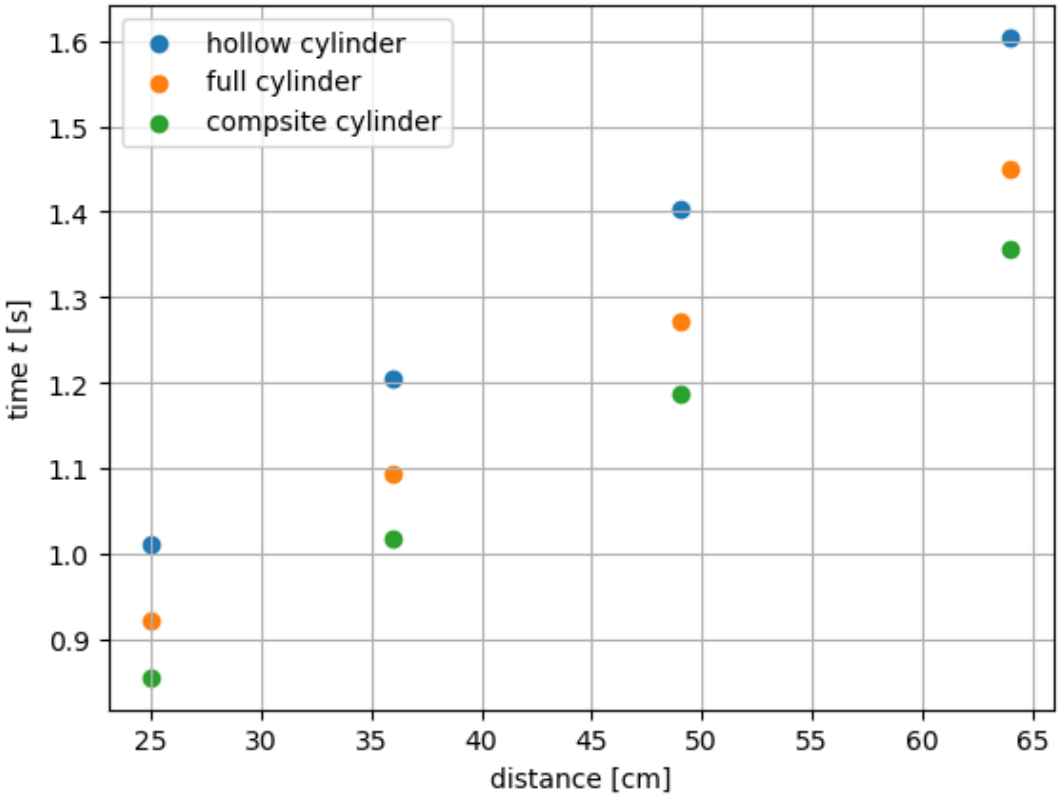
```
12.180000000000001
0.3652396473549936
0.3686461718233353
```

[83]:
```
t_hollow = np.array([1.011, 1.204, 1.403, 1.604])
t_full = np.array([0.921, 1.094, 1.272, 1.451])
t_compose = np.array([0.855, 1.018, 1.186, 1.356])

x = np.array([25, 36, 49, 64])

plt.scatter(x, t_hollow, label="hollow cylinder")
plt.scatter(x, t_full, label="full cylinder")
plt.scatter(x, t_compose, label="compsite cylinder")
plt.xlabel("distance [cm]")
plt.ylabel("time $t$ [s]")
plt.grid(True)
plt.legend()
```

[83]:
```
<matplotlib.legend.Legend at 0x1520a7d90>
```



Quantitative A3:

[84]:
```
x = np.array([25, 36, 49, 64])

t_hollow = np.array([
    [1.019, 1.212, 1.410, 1.610],
    [1.027, 1.223, 1.426, 1.629],
    [1.014, 1.209, 1.410, 1.613],
    [1.022, 1.215, 1.415, 1.614],
    [1.009, 1.201, 1.400, 1.600]
])
t_full = np.array([
    [0.922, 1.096, 1.278, 1.459],
    [0.893, 1.068, 1.245, 1.425],
    [0.900, 1.075, 1.255, 1.435],
    [0.901, 1.074, 1.254, 1.435],
    [0.913, 1.087, 1.266, 1.446]
])

mean_hollow = np.mean(t_hollow, axis=0)
std_hollow = np.std(t_hollow, axis=0, ddof=1)/np.sqrt(t_hollow.
 ↪shape[0])  # Standard error of the mean
mean_full = np.mean(t_full, axis=0)
```

```
std_full = np.std(t_full, axis=0, ddof=1)/np.sqrt(t_full.shape[0])  #␣
 ↪Standard error of the mean

print("Hollow cylinder:")
for distance, mean, std in zip(x, mean_hollow, std_hollow):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")

print("\nFull cylinder:")
for distance, mean, std in zip(x, mean_full, std_full):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")
```

```
Hollow cylinder:
25 cm: mean = 1.0182 s, std = 0.0031 s
36 cm: mean = 1.2120 s, std = 0.0036 s
49 cm: mean = 1.4122 s, std = 0.0042 s
64 cm: mean = 1.6132 s, std = 0.0047 s

Full cylinder:
25 cm: mean = 0.9058 s, std = 0.0052 s
36 cm: mean = 1.0800 s, std = 0.0050 s
49 cm: mean = 1.2596 s, std = 0.0057 s
64 cm: mean = 1.4400 s, std = 0.0058 s
```

a4

[85]:
```
# messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])

m_diff_h = np.array([])
for i in [m1h, m2h, m3h, m4h, m5h]:
    m_diff_h = np.append(m_diff_h, i[1] - i[0])

m_diff_f = np.array([])
for i in [m1f, m2f, m3f, m4f, m5f]:
    m_diff_f = np.append(m_diff_f, i[1] - i[0])
```

```
err_t = 0.001
print("Hollow cylinder time differences:")
print(m_diff_h)
print("Mean time difference for hollow cylinder:", np.mean(m_diff_h))
print("Standard error for hollow cylinder:", np.std(m_diff_h, ddof=1)/
 ↪len(m_diff_h))
print('Full error for hollow cylinder:', np.sqrt((np.std(m_diff_h,␣
 ↪ddof=1)/len(m_diff_h))**2 + err_t**2))

print("\nFull cylinder time differences:")

print(m_diff_f)
print("Mean time difference for full cylinder:", np.mean(m_diff_f))
print("Standard error for full cylinder:", np.std(m_diff_f, ddof=1)/
 ↪len(m_diff_f))
print('Full error for full cylinder:', np.sqrt((np.std(m_diff_f,␣
 ↪ddof=1)/len(m_diff_f))**2 + err_t**2))
```

```
Hollow cylinder time differences:
[0.27  0.271 0.27  0.27  0.269]
Mean time difference for hollow cylinder: 0.2700000000000001
Standard error for hollow cylinder: 0.00014142135623730964
Full error for hollow cylinder: 0.0010099504938362077

Full cylinder time differences:
[0.243 0.244 0.243 0.244 0.243]
Mean time difference for full cylinder: 0.2434
Standard error for full cylinder: 0.00010954451150102927
Full error for full cylinder: 0.001005982107196743
```

Examination

[86]:
```
import uncertainties as unc
import uncertainties.unumpy as unp
```

[87]:
```
m = unc.ufloat(443, 1) * 1e-3       # mass in kg
d_outer = unc.ufloat(50, 0.05) * 1e-3  # outer diameter in m
d_inner = unc.ufloat(46.5, 0.05) * 1e-3 # inner diameter in m

I_hollow = (1/2) * m * ((d_outer/2)**2 + (d_inner/2)**2)
I_full = (1/2) * m * (d_outer/2)**2

print(f"Hollow cylinder moment of inertia: {I_hollow:.7e} kgm²")
print(f"Solid cylinder moment of inertia: {I_full:.7e} kgm²")

h = unc.ufloat(10.1, 0.4)*1e-2  # height in m
l = unc.ufloat(1.000, 0.005)  # length in m\

varphi = unp.arcsin(h / l)
```

```
varphi_deg = varphi * 180 / np.pi
print(f"Angle of inclination: {varphi_deg:.3f} degrees, {varphi:.4f}␣
 ↪radians")
```

```
Hollow cylinder moment of inertia: (2.5817209+/-0.0069469)e-04 kgm²
Solid cylinder moment of inertia: (1.3843750+/-0.0041751)e-04 kgm²
Angle of inclination: 5.797+/-0.232 degrees, 0.1012+/-0.0041 radians
```

[88]:
```
a_solid = (9.81 * unp.sin(varphi)) / (1 + I_full / (m * (d_outer/2)**2))
a_hollow = (9.81 * unp.sin(varphi)) / (1 + I_hollow / (m * (d_outer/
 ↪2)**2))

print(a_solid)
print(a_hollow)
```

```
0.661+/-0.026
0.513+/-0.020
```

Plot the values in Diagram

[89]:
```
mean_hollow_squared = mean_hollow**2
std_hollow_squared = np.sqrt(std_hollow**2 + 0.001**2) * mean_hollow * 2
mean_full_squared = mean_full**2
std_full_squared = np.sqrt(std_full**2 + 0.001**2) * mean_full * 2

s = np.array([25, 36, 49, 64])*1e-2



plt.errorbar(s, mean_full_squared, yerr=std_full_squared, fmt='.',␣
 ↪label="Full Cylinder", capsize=3, color='orange')

plt.errorbar(s, mean_hollow_squared, yerr=std_hollow_squared, fmt='.',␣
 ↪label="Hollow Cylinder", capsize=3, color='blue')



func = lambda x, a: (2.0 / a) * x

popt_hollow, pcov_hollow = curve_fit(func, s, mean_hollow_squared,␣
 ↪sigma=std_hollow_squared, absolute_sigma=True)
popt_full, pcov_full = curve_fit(func, s, mean_full_squared,␣
 ↪sigma=std_full_squared, absolute_sigma=True)

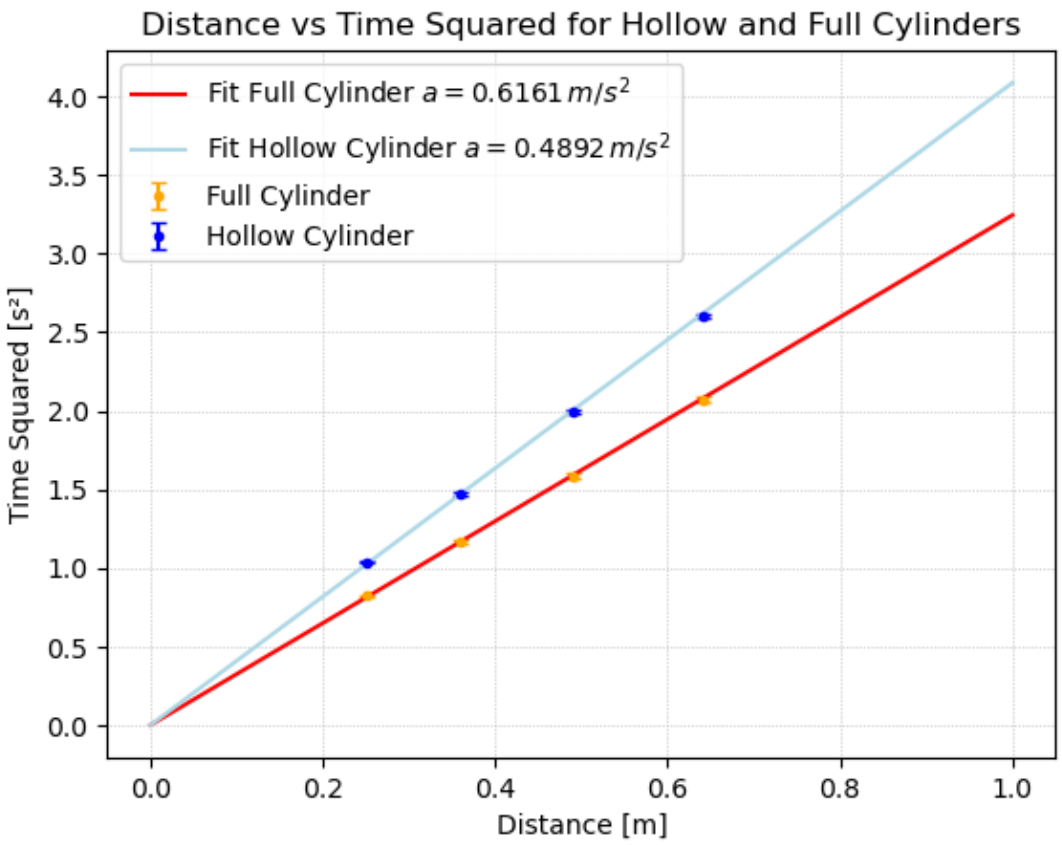
x_fit = np.linspace(0, 1, 100)
plt.plot(x_fit, func(x_fit, popt_full[0]), label=f"Fit Full Cylinder $a␣
 ↪= {popt_full[0]:.4f}\, m/s^2$", color='red')
```

```
plt.plot(x_fit, func(x_fit, popt_hollow[0]), label=f"Fit Hollow␣
 ↪Cylinder $a = {popt_hollow[0]:.4f}\, m/s^2$", color='lightblue')

plt.xlabel("Distance [m]")
plt.ylabel("Time Squared [s²]")
plt.grid(which='both', linestyle=':', linewidth=0.5)
plt.legend()
plt.title("Distance vs Time Squared for Hollow and Full Cylinders")
print(f"Full cylinder fit parameters: slope = {popt_full[0]:.4f}␣
 ↪$\\pm$ {np.sqrt(pcov_full[0][0]):.4f}")
print(f"Hollow cylinder fit parameters: slope = {popt_hollow[0]:.4f}␣
 ↪$\\pm$ {np.sqrt(pcov_hollow[0][0]):.4f}")
```

```
Full cylinder fit parameters: slope = 0.6161 $\pm$ 0.0029
Hollow cylinder fit parameters: slope = 0.4892 $\pm$ 0.0015

<>:22: SyntaxWarning: invalid escape sequence '\,'
<>:23: SyntaxWarning: invalid escape sequence '\,'
<>:22: SyntaxWarning: invalid escape sequence '\,'
<>:23: SyntaxWarning: invalid escape sequence '\,'
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
 ↪668146309.py:22
: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, popt_full[0]), label=f"Fit Full Cylinder␣
 ↪$a =
{popt_full[0]:.4f}\, m/s^2$", color='red')
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
 ↪668146309.py:23
: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, popt_hollow[0]), label=f"Fit Hollow␣
 ↪Cylinder $a =
{popt_hollow[0]:.4f}\, m/s^2$", color='lightblue')
```



Fitted the other way around

[93]:
```
mean_hollow_squared = mean_hollow**2
std_hollow_squared = np.sqrt(std_hollow**2 + 0.001**2) * mean_hollow * 2
mean_full_squared = mean_full**2
std_full_squared = np.sqrt(std_full**2 + 0.001**2) * mean_full * 2

s = np.array([25, 36, 49, 64]) * 1e-2

plt.errorbar(mean_full_squared, s, xerr=std_full_squared, fmt='.',␣
 ↪label='Full Cylinder', capsize=3, color='orange')
plt.errorbar(mean_hollow_squared, s, xerr=std_hollow_squared, fmt='.',␣
 ↪label='Hollow Cylinder', capsize=3, color='blue')

# keep the same model form y = a*x + c, but fix c = 0
func = lambda x, a, c=0: (a / 2.0) * x + c

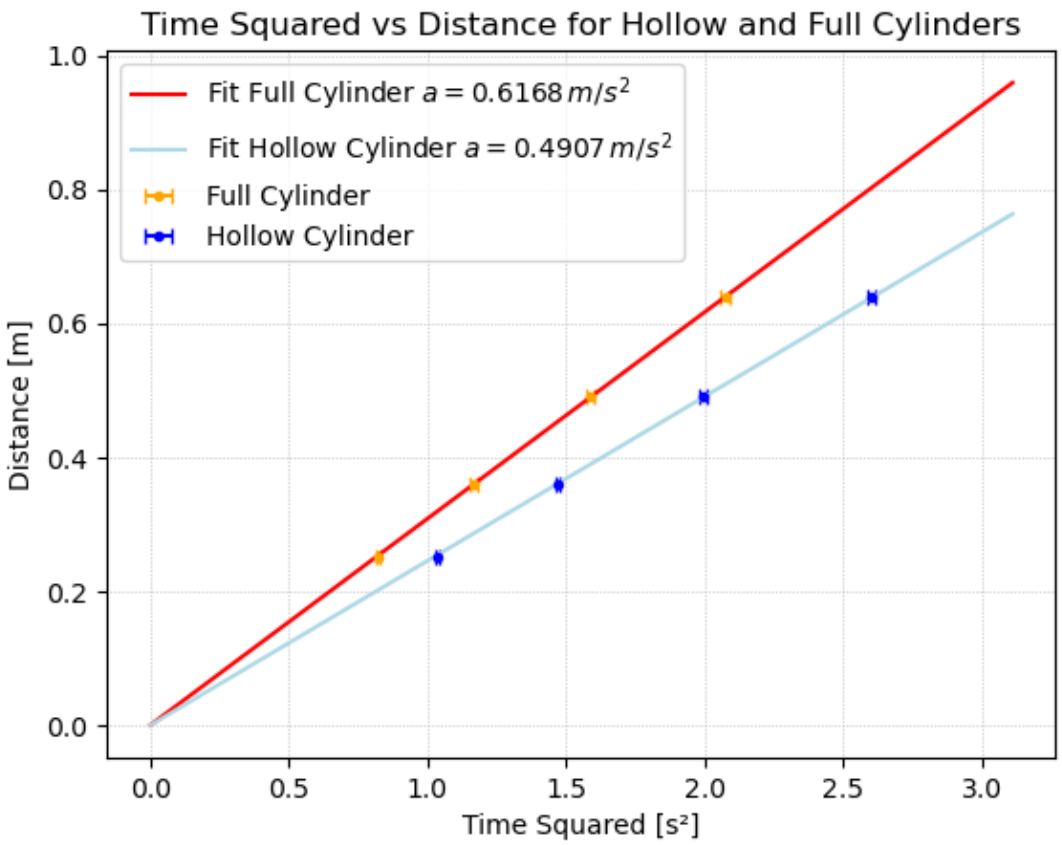
popt_hollow, pcov_hollow = curve_fit(func, mean_hollow_squared, s,␣
 ↪sigma=np.full_like(s, 0.005), absolute_sigma=True, p0=[1.0],␣
 ↪bounds=([-np.inf], [np.inf]))
popt_full, pcov_full = curve_fit(func, mean_full_squared, s, sigma=np.
 ↪full_like(s, 0.005), absolute_sigma=True, p0=[1.0], bounds=([-np.
 ↪inf], [np.inf]))
```

```
x_fit = np.linspace(0, np.max(mean_full_squared) * 1.5, 100)
plt.plot(x_fit, func(x_fit, *popt_full), label=f'Fit Full Cylinder $a =␣
 ↪{popt_full[0]:.4f}\, m/s^2$', color='red')
plt.plot(x_fit, func(x_fit, *popt_hollow), label=f'Fit Hollow Cylinder␣
 ↪$a = {popt_hollow[0]:.4f}\, m/s^2$' , color='lightblue')

plt.xlabel('Time Squared [s²]')
plt.ylabel('Distance [m]')
plt.grid(which='both', linestyle=':', linewidth=0.5)
plt.legend()
plt.title('Time Squared vs Distance for Hollow and Full Cylinders')
print(f'Full cylinder fit parameters: a = {popt_full[0]:.4f} $\\pm$␣
 ↪{np.sqrt(pcov_full[0][0]):.4f}, c = 0')
print(f'Hollow cylinder fit parameters: a = {popt_hollow[0]:.4f}␣
 ↪$\\pm$ {np.sqrt(pcov_hollow[0][0]):.4f}, c = 0')
```

```
<>:18: SyntaxWarning: invalid escape sequence '\,'
<>:19: SyntaxWarning: invalid escape sequence '\,'
<>:18: SyntaxWarning: invalid escape sequence '\,'
<>:19: SyntaxWarning: invalid escape sequence '\,'
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
 ↪3647935948.py:1
8: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, *popt_full), label=f'Fit Full Cylinder $a =
{popt_full[0]:.4f}\, m/s^2$', color='red')
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
 ↪3647935948.py:1
9: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, *popt_hollow), label=f'Fit Hollow Cylinder␣
 ↪$a =
{popt_hollow[0]:.4f}\, m/s^2$' , color='lightblue')

Full cylinder fit parameters: a = 0.6168 $\pm$ 0.0034, c = 0
Hollow cylinder fit parameters: a = 0.4907 $\pm$ 0.0027, c = 0
```



Calculate Height

[ ]:
```
d = unc.ufloat(11, 0.5)*1e-3  # dist start in m

h = h/l * (l-d)
print('delta h in m:', h)
```

```
delta h in m: 0.100+/-0.004
```

[ ]:
```
E = m * 9.81 * h
E
```

[ ]:
```
0.43410061287+/-0.01722142256704715
```

[ ]:
```
m_diff_f
err_f = np.sqrt((np.std(m_diff_f, ddof=1)/len(m_diff_f))**2 + err_t**2)

m_diff_h
err_h = np.sqrt((np.std(m_diff_h, ddof=1)/len(m_diff_h))**2 + err_t**2)

d = unc.ufloat(25, 0.07)*1e-2  # dist between light barriers in m

t_f = unc.ufloat(np.mean(m_diff_f), err_f)
```

```
t_h = unc.ufloat(np.mean(m_diff_h), err_h)


v_f = d / t_f
v_h = d / t_h

r = d_outer / 2


print(f"Full cylinder speed: {v_f:.4f} m/s")
print(f"Hollow cylinder speed: {v_h:.4f} m/s")

w_f = v_f / r
print(f"Full cylinder angular velocity: {w_f:.4f} rad/s")
w_h = v_h / r
print(f"Hollow cylinder angular velocity: {w_h:.4f} rad/s")


E_trans_f = 1/2 * m * v_f**2
E_rot_f = 1/2 * I_full * w_f**2

E_trans_h = 1/2 * m * v_h**2
E_rot_h = 1/2 * I_hollow * w_h**2

print(f"\nFull cylinder translational kinetic energy: {E_trans_f:.5f}␣
 ↪J")
print(f"Full cylinder rotational kinetic energy: {E_rot_f:.5f} J")
print(f"total kinetic energy: {E_trans_f + E_rot_f:.5f} J")
print(f"\nHollow cylinder translational kinetic energy: {E_trans_h:.
 ↪5f} J")
print(f"Hollow cylinder rotational kinetic energy: {E_rot_h:.5f} J")
print(f"total kinetic energy: {E_trans_h + E_rot_h:.5f} J")
```

```
Full cylinder speed: 1.0271+/-0.0051 m/s
Hollow cylinder speed: 0.9259+/-0.0043 m/s
Full cylinder angular velocity: 41.0846+/-0.2092 rad/s
Hollow cylinder angular velocity: 37.0370+/-0.1770 rad/s

Full cylinder translational kinetic energy: 0.23368+/-0.00239 J
Full cylinder rotational kinetic energy: 0.11684+/-0.00120 J
total kinetic energy: 0.35051+/-0.00359 J

Hollow cylinder translational kinetic energy: 0.18990+/-0.00183 J
Hollow cylinder rotational kinetic energy: 0.17707+/-0.00172 J
total kinetic energy: 0.36697+/-0.00354 J
```

[ ]: